

CAPSTONE PROJECT TITLE APPROVAL FORM

Proponents:

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Capstone Project Title: Avelon

Project Type:

- ☒ System Development
- ☒ Mobile Application
- ☐ Game Development
- ☐ Cybersecurity Framework
- ☐ Network Infrastructure

Evaluation

	Approved	Disapproved
Feasibility		
Relevance to the Program		
Originality and Innovation		
Alignment with SDGs		

Overall Recommendation

- ☐ Approved
- ☐ Disapproved

Instructions for re-submission (if disapproved)

Recommending Approval:

Approved by:

Dean Aristotle B. Liwanag

PROJECT INFORMATION	
RESEARCH TITLE	Avelon: Decentralized Lending Platform Enhanced with Blockchain for Automated Risk Assessment, Smart Contract-Based Repayment Enforcement, and Real-Time Ethereum Volatility Prediction
TECHNICAL DESCRIPTION	
RATIONALE	In current decentralized finance (DeFi) ecosystems, lending platforms often suffer from high default risks due to market volatility and a lack of rigorous borrower vetting. Lenders are forced to rely solely on heavy over-collateralization, while borrowers face unpredictable liquidation events without transparent risk analysis. However, a professional and secure crypto lending environment demands: ● Predictive risk modeling for Ethereum volatility ● Strict financial requirement enforcement ● Immutable loan agreement records ● Automated collateral liquidation ● Tamper-proof credit scoring Why Blockchain Is Necessary Blockchain acts as the immutable trust layer that enforces the financial logic defined by the AI risk models. It ensures that every step of the loan lifecycle is: ● Verifiable ● Immutable ● Secure ● Transparent This protects borrowers, and platform administrators against fraud, unauthorized changes to loan terms, and disputes over repayment history. SDG Alignment ● This project directly supports SDG 8 (Decent Work & Economic Growth) by promoting stable, secure financial systems through reduced lending risk and increased trust. It contributes to economic productivity and technological upgrading in finance. ● Additionally, it aligns with SDG 9 (Industry, Innovation, and Infrastructure) by integrating advanced AI risk modeling with decentralized blockchain technology, creating a modern, resilient financial infrastructure.

OBJECTIVES	• To develop a secure, full-stack crypto lending platform integrating a Python/Django backend with a React/Next.js frontend. • To design and implement a novel AI risk assessment model (using LSTM/GRU) that achieves over 85% accuracy in predicting Ethereum price volatility and generates a quantitative borrower eligibility score. • To develop and deploy robust Ethereum Smart Contracts that automate the enforcement of strict lending rules, collateral management (150% minimum ratio), and instant liquidation upon failure to meet repayment terms.										
METHODOLOGY	1. Requirement Analysis & Design (System Design / UI/UX, Architecture): Define all REST API endpoints, finalize the Smart Contract logic, select the architecture for the AI model (LSTM/GRU), and create the overall UI/UX wireframes for the Borrower and Admin interfaces. 2. AI Model Development & Training: Gather historical ETH price data and simulated repayment data. Clean the data and train the LSTM/GRU time-series model. Integrate the trained model with the Django backend for real-time scoring. 3. Smart Contract Development & Audit: Write and test the Solidity Smart Contracts for collateral deposit, loan disbursement, repayment, and automated liquidation. Deploy contracts on an Ethereum test network (e.g., Sepolia) and perform internal security auditing. 4. Full-Stack Implementation: Develop the Django REST API endpoints and integrate them with the deployed Smart Contracts. Build the React frontend to allow users to apply for loans, view AI scores, deposit ETH, and monitor loan status (real-time monitoring). 5. Testing & Evaluation: Conduct comprehensive unit and integration tests (PyTest, Jest). Stress-test the Smart Contracts on the testnet (Blockchain testing on testnet/private network). Perform system performance analysis and evaluate the AI model's prediction accuracy (Evaluation).										
REFERENCES	<table border="1"> <tr> <td data-bbox="342 1297 415 1392">[1]</td><td data-bbox="415 1297 1539 1392">Adamyk, B., Benson, V., Adamyk, O., & Liashenko, O. (2025). <i>Risk management in DeFi: Analyses of the innovative tools and platforms for tracking DeFi transactions</i>. MDPI Journal of Risk and Financial Management. https://www.mdpi.com/1911-8074/18/1/38</td></tr> <tr> <td data-bbox="342 1392 415 1486">[2]</td><td data-bbox="415 1392 1539 1486">Ethereum Foundation. (2024). Ethereum developer documentation: Smart contracts and decentralized applications. ethereum.org. https://ethereum.org/en/developers/docs/</td></tr> <tr> <td data-bbox="342 1486 415 1612">[3]</td><td data-bbox="415 1486 1539 1612">Kaplan, B., Benli, V. F., & Alp, E. A. (2023). Blockchain based decentralized lending protocols: A return analysis between S&P 500 and DeFi assets. <i>Journal of Emerging Economies and Policy</i>, 8(1), 360-378. https://dergipark.org.tr/en/download/article-file/2989413</td></tr> <tr> <td data-bbox="342 1612 415 1707">[4]</td><td data-bbox="415 1612 1539 1707">Infoteks. (2025). LSTM network application for forecasting Ethereum price changes and trends. <i>Journal of Information Systems and Technology</i>. https://infoteks.org/journals/index.php/isikti/article/view/196</td></tr> <tr> <td data-bbox="342 1707 415 1827">[5]</td><td data-bbox="415 1707 1539 1827">SmartDev. (2024). How AI is revolutionizing credit scoring and financial risk assessment. SmartDev Blog. https://smartdev.com/how-ai-is-revolutionizing-credit-scoring/</td></tr> </table>	[1]	Adamyk, B., Benson, V., Adamyk, O., & Liashenko, O. (2025). <i>Risk management in DeFi: Analyses of the innovative tools and platforms for tracking DeFi transactions</i> . MDPI Journal of Risk and Financial Management. https://www.mdpi.com/1911-8074/18/1/38	[2]	Ethereum Foundation. (2024). Ethereum developer documentation: Smart contracts and decentralized applications. ethereum.org. https://ethereum.org/en/developers/docs/	[3]	Kaplan, B., Benli, V. F., & Alp, E. A. (2023). Blockchain based decentralized lending protocols: A return analysis between S&P 500 and DeFi assets. <i>Journal of Emerging Economies and Policy</i> , 8(1), 360-378. https://dergipark.org.tr/en/download/article-file/2989413	[4]	Infoteks. (2025). LSTM network application for forecasting Ethereum price changes and trends. <i>Journal of Information Systems and Technology</i> . https://infoteks.org/journals/index.php/isikti/article/view/196	[5]	SmartDev. (2024). How AI is revolutionizing credit scoring and financial risk assessment. SmartDev Blog. https://smartdev.com/how-ai-is-revolutionizing-credit-scoring/
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EXPECTED OUTPUT	The project will produce: ● A Fully Working Crypto Lending Platform (Mobile Interface for Borrower and Web Interface for Admin(Lender)). ● AI Risk Assessment Model for borrower scoring and ETH volatility prediction, integrated with the backend API. ● Ethereum Smart Contract Suite for automated collateral deposit, loan disbursement, and enforced liquidation. ● Comprehensive Documentation of the system architecture, Smart Contract logic, and AI model training parameters. ● User Manual and Guide for both admin and Borrowers on platform usage and Web3 wallet integration. ● Test Results demonstrating the accuracy of the AI model and the security/functionality of the deployed Smart Contracts.
POTENTIAL IMPACT	● Reduced Default Risk: Avelon mitigates uncertainty in crypto lending by mandating strict collateral requirements and implementing AI-vetted borrower scores, creating a more predictable investment environment for the admin(lender). ● Promotes Stable Economic Growth (SDG 8): By introducing a data-driven, accountable, and transparent financial mechanism, the platform fosters trust and stability, which is essential for economic participation. ● Elevated DeFi Standards: The integration of professional risk modeling and automated contract enforcement elevates the entire decentralized finance industry toward a more regulated-like and secure structure.
USERS OR BENEFICIARIES	● Clearly identify who will directly use or benefit from your project. ● Borrowers: Qualified users gain secure, transparent access to funds using their crypto collateral without needing a traditional bank or middleman. ● Financial Analysts & DeFi Investors: Benefit from access to a new, structured platform that utilizes innovative AI risk modeling for research and investment exploration. ● Platform Admins/Compliance Advisors: Benefit from automated enforcement, centralized risk threshold management, and clear platform KPIs.